

Firmware

The program for the microcontroller is written in Assembly language, in serial interrupt mode. It is a simple program to test and study the working of serial COM port of a PC, and serially interface it with a microcontroller. It has the code for sending serial data and displaying the received signal. The Assembly program code (SIJ.asm) is listed at the bottom of this page.

Construction and testing

An actual-size, single-side PCB layout of the circuit for serial port interfacing with LEDs in Java is shown in Fig. 5 and its component layout in Fig. 6.

Step-by-step procedure for testing the project follows:

1. Compile the Java codes using JCreator.

2. Compile the Assembly code using a suitable software such as ASM51 cross-assembler. Burn the hex code into the microcontroller chip and then place the chip in the circuit.

3. Set port P1 of microcontroller to FF (or 255) by pulling all the switches S1 through S8 to +5V. This will enable handshaking.

4. Run the Java program. Select COM port. Here COM1 port is used (refer Fig. 4).

5. Set baud rate to 9600, data bit to 8, parity bit to none and stop bit to 1.

6. Now type a value between 0 and 255 in 'Send Data' field.

7. Click 'Connect' button. After sending data for the first time, the software waits for handshaking signal FF (Hex) from the hardware device. If port P1 is set to FF (Hex), the hand-

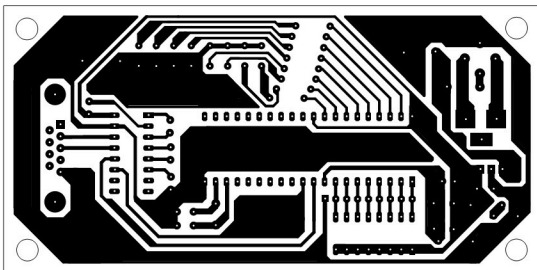


Fig. 5: An actual-size, single-side PCB layout of the circuit for serial port interfacing with LEDs in Java

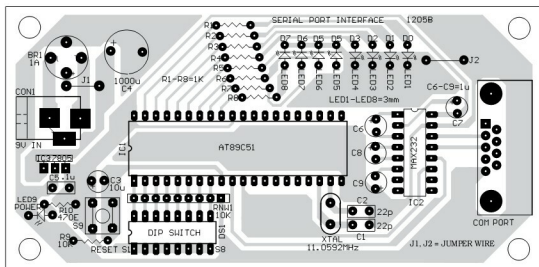


Fig. 6: Component layout for the PCB

shaking will be successful. The data sent from the hardware device will be displayed in 'Received Data' text box on the screen. In this case, FFH is displayed. Any value between 0 and 255 entered in the software is displayed in 'Received Data' text box and also on the LEDs (LED1 through LED8). Similarly, changes made in the input switches should be displayed in 'Received Data' box as well as on the LEDs.

EFY note. 1. The rxtSerial.dll file can be downloaded from:

http://rapidlibrary.com/files/rxtserial-dll_ulz99nee8ci89on.html

2. The download links for gnu/io folder are:

www.jcontrol.org/download/files/rxtx-2.1-7-bins-r2.zip

http://www.jcontrol.org/download/rxtx_en.html

3. The source codes of the project and related files are available at www.efymag.com website.

The author is a software developer at Software Associates, Kozhikode (Calicut)

SIJ.ASM

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ORG 00H
SJMP MAIN
ORG 23H
LJMP SER_INT
ORG 30H
MAIN:  MOV P3,#0FFH
        MOV P2,#00H
        MOV P1,#0FFH
        MOV P0,#00H
        LCALL SER_INIT

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BEGIN:  MOV A,P1
        SJMP SBUF,A
        SJMP BEGIN
SER_INIT: MOV TH0,#20H
        MOV TL0,#0FH
        MOV SCON,#50H
        MOV IE,#90H
        SETB IP.4
        SETB TR1
        RET

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SER_INT: JB TI,TRANS
        PUSH 0E0H
        MOV A,SBUF
        MOV P0,A
        CLR RI
        POP 0E0H
        RETI
        CLR TI
        RETI
END
TRANS:

```